

The empirical recurrence for OEIS A250638, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A250638 records “Empirical recurrence of order 57” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 482$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{57} = -77760 \neq 0$, so no further tail satisfies anything shorter; any order-57 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A250638 is “Number of $(6+1)X(n+1)$ 0..2 arrays with nondecreasing $x(i,j)+x(i,j-1)$ in the i direction and nondecreasing $\min(x(i,j),x(i-1,j))$ in the j direction.”. It has offset 1 and begins

7307, 53847, 381834, 2290189, 12291286, 60472234, 280151655, 1237636447, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:38:10, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 482$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 482$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 57: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 57.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 964$ exact terms from $n = 7$ on, it returns order 57 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	23	c_{16}	3472689173	c_{31}	−1206542801790	c_{46}	15018115811
c_2	−190	c_{17}	8723699804	c_{32}	−3625202647647	c_{47}	31835053414
c_3	353	c_{18}	−25726013518	c_{33}	3484539000003	c_{48}	−13256870078
c_4	4593	c_{19}	−4193704083	c_{34}	1679623302135	c_{49}	−2298703943
c_5	−30949	c_{20}	95864395033	c_{35}	−4130310146275	c_{50}	2860653336
c_6	22082	c_{21}	−85842273563	c_{36}	751060071643	c_{51}	−435476268
c_7	471844	c_{22}	−197770374982	c_{37}	2784423905905	c_{52}	−227087784
c_8	−1700987	c_{23}	415097624111	c_{38}	−1865146320199	c_{53}	95698008
c_9	−1559780	c_{24}	116961943037	c_{39}	−895034355150	c_{54}	−4180896
c_{10}	22135203	c_{25}	−1030368166285	c_{40}	1462766903759	c_{55}	−4630176
c_{11}	−31599663	c_{26}	590504309361	c_{41}	−160220198502	c_{56}	1079568
c_{12}	−124352143	c_{27}	1482237959106	c_{42}	−614111283567	c_{57}	−77760
c_{13}	452137420	c_{28}	−2106928078091	c_{43}	306380571478		
c_{14}	84084665	c_{29}	−853377727359	c_{44}	114314919781		
c_{15}	−2788182229	c_{30}	3615944373063	c_{45}	−142336579854		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A250638 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{57} - c_1 x^{56} - \dots - c_{57}$. Its constant term is $-c_{57} = 77760$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 57 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 57, so $\deg q = 57$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 57, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 57, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -77760 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-57 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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